

QUANTUM MECHANICS | ADVANCED UNDERGRADUATE PHYSICS

Sequential Stern–Gerlach Measurements

Quantum Measurement Probabilities and Basis Transformations for Spin-1 Systems

Author: Robbie Kohn

Level: Junior/Senior Undergraduate
(PHY 341) | 50–60 Minutes

Prereqs: Spin-1/2 Stern–Gerlach,
Matrix Mechanics, Dirac Notation,
Born Rule

Artifact: Instructional Sample

Central Idea: Demonstrating that quantum measurement probabilities arise from basis projections, and confronting the breakdown of spin-1/2 geometric intuition in three-level systems.

Instructional Focus & Learning Objectives

In introductory quantum mechanics, students build their foundational intuition of measurement collapse using the two-state spin-1/2 system (Silver atoms). However, the spin-1/2 Hilbert space is deceptively simple: perpendicular spatial measurements (e.g., measuring S_z on an S_x eigenstate) always yield a symmetric 50–50 probability split due to the unique geometry of the Bloch sphere.

This instructional plan guides advanced physics students into the richer, three-dimensional Hilbert space of **spin-1 particles**. By analyzing sequential Stern–Gerlach analyzers with rotated field axes, students discover how probability amplitudes are computed through formal basis projections, confronting and dismantling intuitive misconceptions inherited from spin-1/2.

- **Conceptual Objective:** Understand that changing quantum measurement bases is mathematically analogous to rotating coordinate systems: the physical quantum state remains invariant while its expansion coefficients (amplitudes) transform.
- **Analytical Objective:** Construct the matrix representation of the angular momentum operator $\hat{S}_y$ in the standard $\hat{S}_z$ eigenbasis using ladder operator relations.
- **Procedural Objective:** Solve the matrix eigenvalue problem to express an arbitrary $\hat{S}_y$ eigenstate $|1, \mu\rangle_y$ as a linear combination of $\hat{S}_z$ basis kets $\{|1, 1\rangle, |1, 0\rangle, |1, -1\rangle\}$.
- **Physical Interpretation:** Apply the Born Rule $P(S_z = m\hbar) = |\langle 1, m | 1, \mu \rangle_y|^2$ to calculate measurement probabilities, interpreting why a beam prepared in $S_y = 0$ yields a complete node (0% probability) in the $S_z = 0$ channel.

Anticipated Misconceptions & Pedagogical Guidance

Central Misconception: Equal Probability Fallacy for Perpendicular Measurements

Student Assumption: “For a spin-1/2 particle, measuring along the z-axis after preparing the state along the perpendicular x-axis always gives equal probabilities ($P = 1/2$ for each outcome). Therefore, for a spin-1 particle with three outcomes, measuring along z after preparing the state along the perpendicular y-axis must split the beam equally into thirds ($1/3, 1/3, 1/3$).”

Physical & Mathematical Reality: This intuitive extrapolation is completely false. While the spatial

magnetic field axes are perpendicular in $\mathbb{R}^3$, the state space is a 3-dimensional complex Hilbert space $\mathbb{C}^3$ where state vectors are governed by the spin-1 representation of $\text{SO}(3)$. The probability amplitudes depend on the specific magnetic quantum numbers ($m = 1, 0, -1$). As derived below, for the state $|1, 0\rangle_y$, the probabilities along S_z are:

$$P(S_z = +\hbar) = \frac{1}{2}, \quad P(S_z = 0) = 0, \quad P(S_z = -\hbar) = \frac{1}{2}$$

The center channel is completely extinguished! Far from an even third, the middle outcome has zero probability.

Pedagogical Intervention: Use this unexpected result as a teaching hook. Emphasize that in quantum mechanics, probabilities cannot be guessed from spatial symmetry alone; they must be rigorously calculated via inner products of basis states.

Lesson Launch: Beyond the Two-Level System

Teacher Guidance & Pacing: Motivation (3–5 Minutes)

What to Say to the Class:

“Over the past two weeks, we explored spin-1/2 particles. We saw that if we prepare an electron with spin pointing in the $+x$ direction ($|+_x\rangle$), and then pass it through a Stern–Gerlach apparatus oriented along the z -axis (SG_z), the beam splits exactly 50/50 into $S_z = +\hbar/2$ and $S_z = -\hbar/2$. This feels intuitive: the x -axis lies symmetrically halfway between $+z$ and $-z$.

Today, we step into the physics of **spin-1 particles**—such as neutral mesons or atomic nuclei. In a Stern–Gerlach magnetic field with an inhomogeneous gradient $\frac{\partial B_z}{\partial z}$, a spin-1 beam splits into *three* distinct channels: $S_z = +\hbar$, $S_z = 0$, and $S_z = -\hbar$.

What happens if we prepare a particle in a definite state along the y -axis, say $S_y = 0$, and then send it through an SG_z analyzer? Does it split evenly into thirds? Today we will compute the exact probabilities and discover why our geometric intuition from spin-1/2 breaks down.”

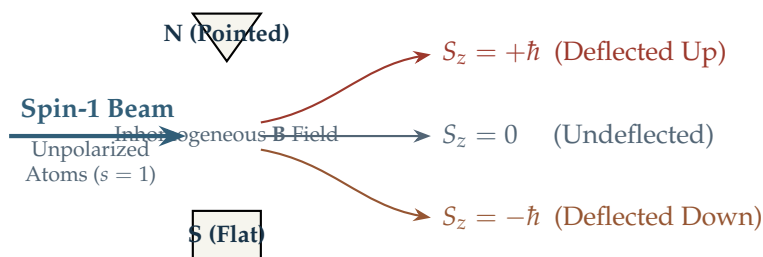


Figure 1: A Stern–Gerlach device splitting a spin-1 beam into three distinct components.

Key Concept: Quantum Measurement as Basis Projection

Measurement probabilities in quantum mechanics arise from the projection of the prepared physical state onto the eigenstates of the measurement operator.

The Fundamental Measurement Postulate (Born Rule)

If a quantum system is prepared in state $|\psi_{\text{prep}}\rangle$, the probability of obtaining eigenvalue a_n upon measuring observable $\hat{A}$ (with orthonormal eigenstates $\{|a_n\rangle\}$) is given by:

$$P(A = a_n) = |\langle a_n | \psi_{\text{prep}} \rangle|^2 = |\text{Probability Amplitude}|^2$$

For sequential Stern–Gerlach measurements where an SG_y filter selects the state $|1, \mu\rangle_y$ and an SG_z device measures S_z :

$$P(S_z = m\hbar) = |{}_z\langle 1, m | 1, \mu \rangle_y|^2 \quad (m, \mu \in \{1, 0, -1\})$$

Conceptual Analogy: Changing Coordinate Bases

What to Say:

“Think of this process like rotating coordinate systems in classical physics. A physical displacement vector $\mathbf{v}$ exists independently of how we draw our axes. If we write $\mathbf{v}$ in a rotated coordinate system (x', y', z') , the physical vector does not change, but its scalar components $(v_{x'}, v_{y'}, v_{z'})$ do. In quantum mechanics, the prepared state $|1, \mu\rangle_y$ is the physical vector. The SG_z measurement apparatus defines our coordinate axes $\{|1, 1\rangle, |1, 0\rangle, |1, -1\rangle\}$. To predict the probability of finding the particle in any given channel, we must resolve the prepared vector along the measurement axes by computing the inner product $\langle 1, m | 1, \mu \rangle_y$.”

The Sequential Stern–Gerlach Apparatus

Consider the sequential experiment illustrated below:

1. A beam of spin-1 particles passes through an initial Stern–Gerlach device oriented along the y -axis (SG_y).
2. The SG_y device separates the beam into three channels: $S_y = +\hbar$, $S_y = 0$, and $S_y = -\hbar$.
3. We place an aperture that blocks the top and bottom beams, transmitting **only** the middle beam: the state with definite eigenvalue $S_y = 0$ ($|1, 0\rangle_y$).
4. This filtered beam enters a second device oriented along the z -axis (SG_z).

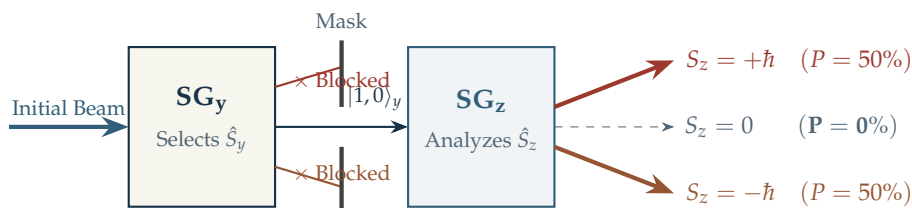


Figure 2: Sequential measurement setup: filtering $S_y = 0$, followed by an S_z Stern–Gerlach analyzer.

Mathematical Development: Matrix Representation of $\hat{S}_y$

To calculate the projection of $|1, 0\rangle_y$ onto the S_z basis kets, we must find the eigenvectors of the spin-1 operator $\hat{S}_y$ expressed in the standard S_z representation.

In the S_z basis $\{|1, 1\rangle, |1, 0\rangle, |1, -1\rangle\}$, the operator $\hat{S}_z$ is diagonal by definition:

$$\hat{S}_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

Using the angular momentum ladder operators $\hat{S}_{\pm} = \hat{S}_x \pm i\hat{S}_y$ with matrix elements $\langle s, m \pm 1 | \hat{S}_{\pm} | s, m \rangle = \hbar \sqrt{s(s+1) - m(m \pm 1)}$, the matrix representation of $\hat{S}_y = \frac{1}{2i}(\hat{S}_+ - \hat{S}_-)$ for $s = 1$ is:

$$\hat{S}_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix} \quad (1)$$

Pedagogical Note: Structure of the Matrix

Point out to students:

- The matrix is Hermitian ($\hat{S}_y^\dagger = \hat{S}_y$), guaranteeing real eigenvalues.
- The diagonal elements are zero because $\langle 1, m | \hat{S}_y | 1, m \rangle = 0$: a state with definite S_z has an expectation value of zero for transverse spin.
- The overall factor $\frac{\hbar}{\sqrt{2}}$ ensures that the characteristic eigenvalues of $\hat{S}_y$ are the required physical values: $+\hbar, 0, -\hbar$.

Worked Example: Determining Channel Fractions for $S_y = 0$

Problem Statement (Townsend, Example 3.5)

A beam of spin-1 particles exiting an SG_y filter in the $S_y = 0$ state enters an SG_z apparatus. Determine the fraction of particles that exit in each of the three S_z channels ($S_z = +\hbar$, $S_z = 0$, and $S_z = -\hbar$).

Step 1: Set up the matrix eigenvalue equation.

Let the state $|1, 0\rangle_y$ be represented in the S_z basis as a column vector:

$$|1, 0\rangle_y \longleftrightarrow \begin{pmatrix} a \\ b \\ c \end{pmatrix} \quad \text{where } a = \langle 1, 1 | 1, 0 \rangle_y, \quad b = \langle 1, 0 | 1, 0 \rangle_y, \quad c = \langle 1, -1 | 1, 0 \rangle_y$$

The eigenvalue equation is $\hat{S}_y|1,0\rangle_y = 0 \cdot \hbar|1,0\rangle_y = 0$:

$$\frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Step 2: Solve the linear system.

Multiplying the matrix by the vector yields the homogeneous system:

$$-ib = 0 \quad (2)$$

$$ia - ic = 0 \quad (3)$$

$$ib = 0 \quad (4)$$

From Equation (2) (or (4)), we obtain immediately:

$$b = 0$$

From Equation (3), dividing by i :

$$a - c = 0 \implies a = c$$

Therefore, any eigenvector with eigenvalue 0 must have the form:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} a \\ 0 \\ a \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

Step 3: Normalize the state vector.

The total probability must sum to unity:

$$|a|^2 + |b|^2 + |c|^2 = |a|^2 + 0 + |a|^2 = 2|a|^2 = 1 \implies |a| = \frac{1}{\sqrt{2}}$$

Choosing the global phase convention where a is real and positive ($a = \frac{1}{\sqrt{2}}$):

$$|1,0\rangle_y \longleftrightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad (5)$$

Step 4: Express in Dirac ket notation.

Reconstructing the state in the $\hat{S}_z$ basis:

$$|1,0\rangle_y = \frac{1}{\sqrt{2}}|1,1\rangle + 0|1,0\rangle + \frac{1}{\sqrt{2}}|1,-1\rangle \quad (6)$$

Step 5: Compute measurement probabilities using the Born Rule.

$$P(S_z = +\hbar) = |\langle 1,1|1,0\rangle_y|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \quad (50\%)$$

$$P(S_z = 0) = |\langle 1,0|1,0\rangle_y|^2 = |0|^2 = 0 \quad (0\%)$$

$$P(S_z = -\hbar) = |\langle 1,-1|1,0\rangle_y|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \quad (50\%)$$

Physical Interpretation & Instructional Closure

What to Emphasize in Class Discussion (5 Minutes)

The Dramatic Physical Result:

“Look at what happened to the middle channel. We sent particles with $S_y = 0$ into an analyzer measuring S_z . Not a single particle emerges with $S_z = 0$! The entire beam splits equally (50% and 50%) into the $S_z = +\hbar$ and $S_z = -\hbar$ beams.

Why? Because in the state $|1, 0\rangle_y$, the probability amplitude for the $|1, 0\rangle$ basis state is exactly zero ($b = 0$). Destructive quantum interference completely cancels the component along the unperturbed beam line.

Now contrast this with the case where the initial beam was prepared in $S_y = +\hbar$ ($|1, 1\rangle_y$):

$$|1, 1\rangle_y = \frac{1}{2}|1, 1\rangle + \frac{i}{\sqrt{2}}|1, 0\rangle - \frac{1}{2}|1, -1\rangle$$

For $S_y = +\hbar$, the middle channel probability is $\left|\frac{i}{\sqrt{2}}\right|^2 = \frac{1}{2}$ (50%), while the top and bottom channels each get $\left|\frac{1}{2}\right|^2 = \frac{1}{4}$ (25%).

The Big Takeaway: In a three-level quantum system, perpendicular measurements do not exhibit equal splits. The distribution of measurement outcomes depends intimately on the quantum state's projection in Hilbert space.”

Formative Assessment & Individual Check

Name: _____ Date: _____

Task 1 (Eigenstate Derivation): Starting from the matrix representation of $\hat{S}_y$, verify that the normalized state for the top beam $S_y = +\hbar$ is:

$$|1, 1\rangle_y = \frac{1}{2} \begin{pmatrix} 1 \\ i\sqrt{2} \\ -1 \end{pmatrix}$$

Show the substitution into the linear system and demonstrate that the norm squared equals 1.

Task 2 (Measurement Prediction): If this beam ($|1, 1\rangle_y$) is injected into the SG_z apparatus, calculate the

theoretical percentage of particles detected in each of the three channels. Verify that the sum of the probabilities equals 100%.

Curricular Context & References

This lesson plan is designed for an advanced undergraduate course in Quantum Mechanics (junior/senior level), corresponding to Chapter 3 of Townsend's textbook.

- Townsend, J. S. *A Modern Approach to Quantum Mechanics*, 2nd ed.; University Science Books: Mill Valley, CA, 2012; Chapter 3 (Angular Momentum), Section 3.5 (Spin-1 Particles), pp. 75–82.
- Sakurai, J. J.; Napolitano, J. *Modern Quantum Mechanics*, 3rd ed.; Cambridge University Press, 2020; Chapter 1 (Fundamental Concepts).
- McIntyre, D. H. *Quantum Mechanics: A Paradigms Approach*; Pearson, 2012; Chapter 2.